

WIPRO CAPSTONE PROJECT REPORT	
Name	Jithin Varma
Superset ID	6385860
Organization	Wipro
Project Type	Capstone Project
Project Name	Automation Testing using Selenium PyTest Framework
	Automation Testing using Behave BDD Framework
Domain	E-Commerce / Online Grocery Application
Module Title	BigBasket Pharmacy & Wellness Module

SYNOPSIS

The Wipro Capstone Project focuses on automating the BigBasket Pharmacy & Wellness module using two separate automation frameworks: Selenium WebDriver with Python PyTest framework and Behave BDD framework.

The automation validates user login workflow, pharmacy page navigation, brand filter selection, price filter validation, add-to-basket functionality, basket navigation, checkout workflow, payment page validation, invalid mobile validation, and invalid OTP validation.

The framework was designed using the Page Object Model (POM) architecture to improve maintainability, reusability, and scalability. Screenshots, logs, Allure reports, reusable utilities, CSV data handling, and assertions were integrated to support execution tracking and debugging.

The project includes:

- End-To-End Testing
- Positive Testing
- Negative Testing
- Data-Driven Testing

- Behavior Driven Development (BDD)

using industry-standard automation testing practices.

PROJECT DESCRIPTION

The automation framework automates the complete workflow of the BigBasket Pharmacy & Wellness module. The framework performs user login validation, pharmacy page navigation, brand filter selection, price filter validation, product selection, add-to-basket validation, basket navigation, checkout validation, and payment page verification.

Separate page classes were implemented for Login, Pharmacy, Basket, and Checkout modules using Selenium Python Page Object Model framework to ensure reusable and maintainable automation design.

The project also includes Behave BDD implementation where business scenarios are written using Gherkin syntax and integrated with Selenium automation scripts using Given-When-Then structure.

The framework supports:

- Selenium PyTest Framework
- Selenium Behave BDD Framework
- Allure Reporting
- Screenshot Capture
- Logging Framework
- CSV Data Handling
- Assertions
- Reusable Utilities

The framework provides professional automation reporting, execution tracking, validation handling, and scalable automation architecture for real-world web application testing.

SELENIUM PYTEST FRAMEWORK

TEST STRATEGY

The project uses Selenium WebDriver with Python PyTest framework for automation testing. The framework supports:

- End-to-End Testing
- Positive Testing
- Negative Testing
- Data-Driven Testing

Tools Used:

- Selenium WebDriver
- PyTest
- Allure Reports
- Logging Module

TEST ENVIRONMENT

Operating System	macOS
IDE	PyCharm Community Edition
Browser	Google Chrome
Framework	Selenium WebDriver + PyTest
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports

FRAMEWORK ARCHITECTURE

Project_BigBasket

- |— config
- |— data
- |— logs
- |— pages
- |— screenshots
- |— tests
- |— utils
- |— allure-results
- |— allure-report
- |— conftest.py
- |— pytest.ini
- |— requirements.txt
- └— README.md

TEST EXECUTION

The automation suite includes:

- 1 End-to-End Test Case
- 4 Positive Test Cases
- 2 Negative Test Cases

The End-to-End test validates the complete BigBasket Pharmacy & Wellness workflow from login validation until payment page verification.

S.No	Test Type	Test Scenario	Expected Result	Status
1	End-to-End	Complete BigBasket Pharmacy Workflow	User successfully reaches payment page	Passed
2	Positive	Brand Filter Validation	Brand filter applied successfully	Passed
3	Positive	Price Filter Validation	Price filter applied successfully	Passed
4	Positive	Add To Basket Validation	Product added to basket successfully	Passed
5	Positive	Basket Navigation Validation	Basket page opens successfully	Passed
6	Negative	Invalid Mobile Number Validation	Validation message displayed successfully	Passed
7	Negative	Invalid OTP Validation	Invalid OTP validation handled successfully	Passed

END-TO-END FLOW VALIDATION

The End-to-End automation validates the complete business workflow including:

- User Login
- Pharmacy Page Navigation
- Brand Filter Application
- Price Filter Application
- Product Selection
- Add Product To Basket
- Basket Navigation
- Checkout Navigation
- Payment Page Validation

POSITIVE TEST CASES

Positive Test Case 1 – Brand Filter Validation

- Validates brand filter functionality
 - Selects “Dettol” brand filter
 - Verifies successful brand filter application
 - Checks pharmacy page navigation successfully
-

Positive Test Case 2 – Price Filter Validation

- Validates price filter functionality
 - Selects product price range dynamically
 - Verifies successful price filter application
 - Ensures pharmacy products are filtered correctly
-

Positive Test Case 3 – Add Product To Basket Validation

- Validates add-to-basket functionality
 - Adds pharmacy product successfully
 - Verifies basket update successfully
 - Ensures selected product added correctly
-

Positive Test Case 4 – Basket Navigation Validation

- Validates basket opening functionality
 - Opens basket successfully
 - Verifies successful basket page navigation
 - Ensures added products are visible in basket
-

NEGATIVE TEST CASES

Negative Test Case 1 – Invalid Mobile Number Validation

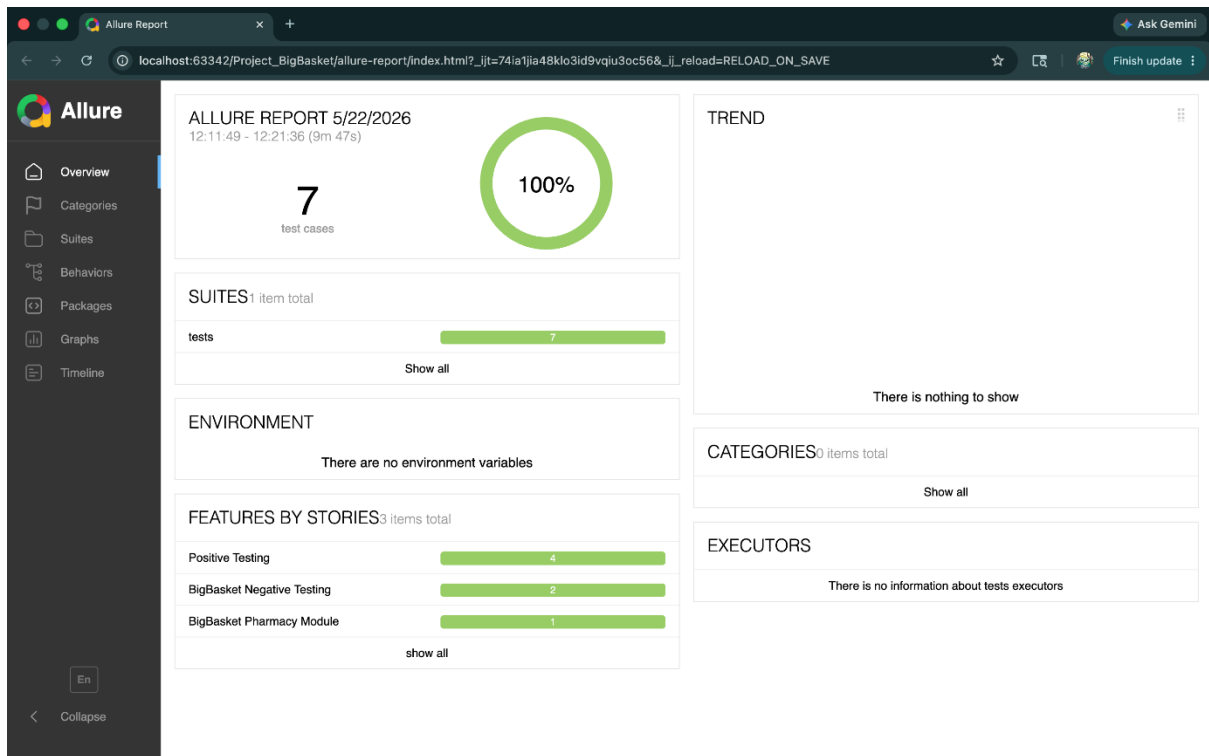
- Enters invalid mobile number
 - Validates application error handling
 - Verifies invalid mobile validation successfully
 - Ensures unauthorized login prevention
-

Negative Test Case 2 – Invalid OTP Validation

- Enters valid mobile number
- Proceeds to OTP verification screen
- Verifies invalid OTP validation handling
- Ensures user login restricted for invalid OTP values

ALLURE REPORTS

- Integrated Allure Reports with PyTest
- Generates professional execution reports
- Displays passed and failed test status
- Captures screenshots during execution
- Maintains execution history
- Displays step-by-step execution flow
- Helps debugging and tracking failures
- Provides detailed execution summary
- Displays assertion validation results
- Provides clean automation execution tracking
- Generates real-time execution analytics
- Improves automation reporting quality

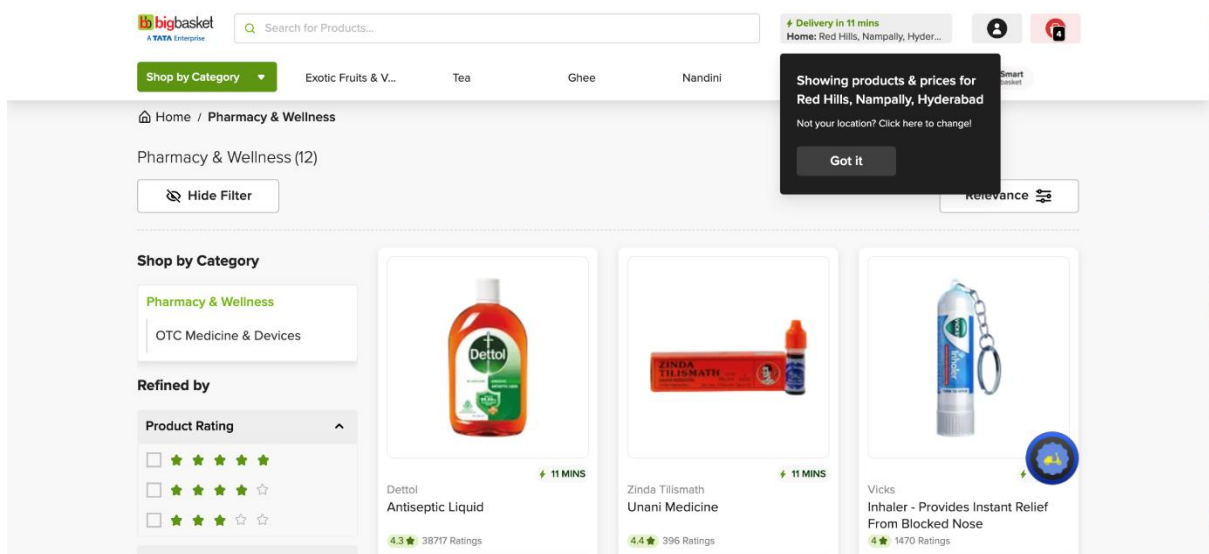
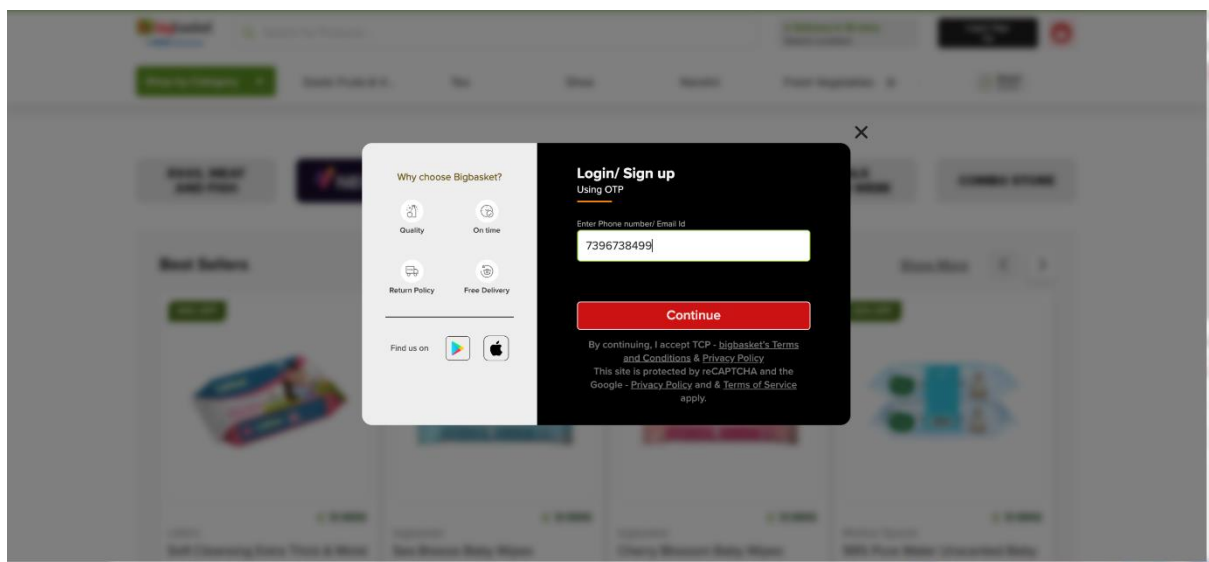
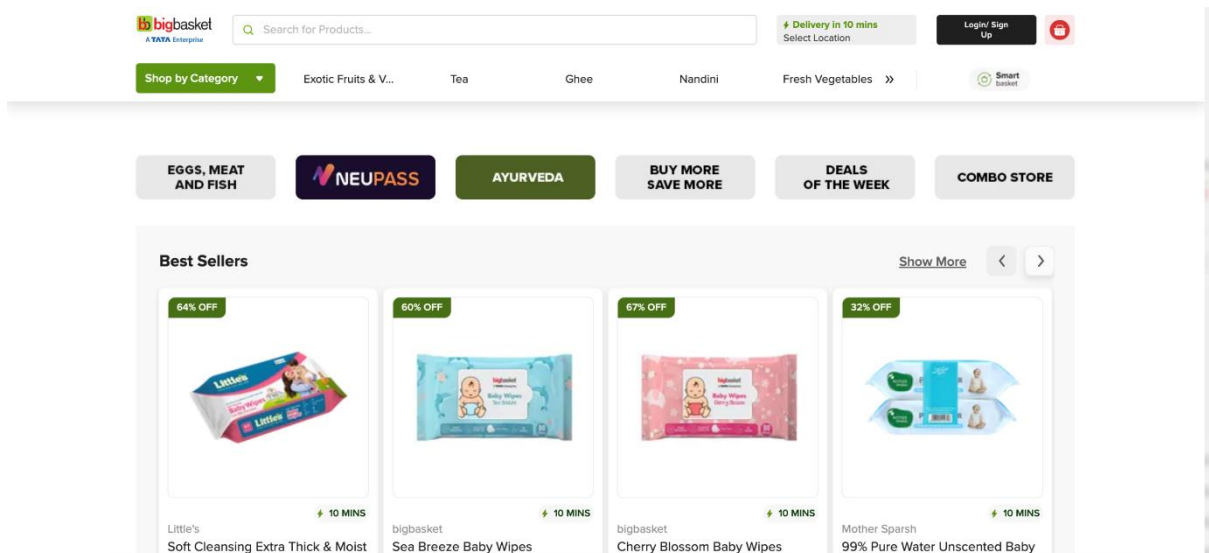


TEST OUTPUTS

The automation framework automatically captures screenshots during important execution stages including user login validation, pharmacy page navigation, brand filter application, price filter validation, add-to-basket functionality, basket navigation, checkout workflow validation, and payment page verification. These screenshots help in execution tracking, validation confirmation, debugging, and maintaining visual proof of automation execution.

Detailed execution logs are generated using the Python logging framework for debugging, execution monitoring, validation tracking, failure analysis, and automation flow verification. The logging framework helps in identifying execution failures, tracking step-by-step automation flow, monitoring assertions, and improving automation maintenance and debugging efficiency.

The framework also generates professional Allure Reports which provide detailed execution summaries including passed and failed test status, execution history, screenshots, assertion validations, step-by-step execution flow, and real-time automation execution analytics. These reports improve reporting quality, automation tracking, and execution transparency for real-world automation testing practices.



TEST REPORT

- ❑ Total Tests Executed: 7
- ❑ Tests Passed: 7
- ❑ Tests Failed: 0
- ❑ Automation Accuracy: 100%

CONCLUSION

The Selenium PyTest automation framework successfully validates the BigBasket Pharmacy & Wellness application using End-To-End Testing, Positive Testing, and Negative Testing approaches.

The framework is designed using:

- Selenium WebDriver
- PyTest Framework
- Page Object Model (POM)
- Allure Reporting
- Logging Framework
- Data-Driven Testing
- Screenshot Capture
- Assertion Validations

The automation framework improves:

- Test Maintainability
- Reusability
- Scalability
- Reporting Quality
- Automation Reliability
- Execution Tracking
- Validation Accuracy

Overall, the project demonstrates a practical, scalable, and professional Selenium automation framework for real-world e-commerce web application testing using industry-standard automation testing practices.

SELENIUM BDD FRAMEWORK

TEST STRATEGY

The project uses Selenium WebDriver with Python Behave BDD framework for automation testing. The framework supports:

- End-To-End Testing
- Positive Testing
- Negative Testing
- Data-Driven Testing
- Behavior Driven Development (BDD)

Tools Used:

- Selenium WebDriver
- Behave BDD
- Allure Reports
- Logging Module
- CSV Data Handling
- Python

TEST ENVIRONMENT

Parameter	Details
Operating System	macOS
IDE	PyCharm Community Edition
Browser	Google Chrome
Framework	Selenium WebDriver + Behave BDD
Design Pattern	Page Object Model (POM)
Reporting Tool	Allure Reports
Test Approach	Behavior Driven Development

FRAMEWORK ARCHITECTURE

BDD_BigBasket
└─ config

```
|— data
|— features
|   |— end_to_end.feature
|   |— positive.feature
|   |— negative.feature
|   |— environment.py
|   |— steps
|       |— end_to_end_steps.py
|       |— positive_steps.py
|       |— negative_steps.py
|— logs
|— pages
|— screenshots
|— utils
|— allure-results
|— allure-report
|— behave.ini
|— requirements.txt
|— README.md
```

TEST EXECUTION

The automation suite includes:

- 1 End-To-End Test Scenario
- 4 Positive Test Scenarios
- 2 Negative Test Scenarios

The End-To-End scenario validates the complete BigBasket Pharmacy & Wellness workflow from user login validation until payment page verification.

S.No	Test Type	Test Scenario	Expected Result	Status
1	End-To-End	Complete BigBasket Pharmacy Workflow	User successfully reaches payment page	Passed

S.No	Test Type	Test Scenario	Expected Result	Status
2	Positive	Brand Filter Validation	Brand filter applied successfully	Passed
3	Positive	Price Filter Validation	Selected price filter applied successfully	Passed
4	Positive	Add Product To Basket Validation	Product added to basket successfully	Passed
5	Positive	Basket Navigation Validation	Basket page opens successfully	Passed
6	Negative	Invalid Mobile Number Validation	Validation message displayed successfully	Passed
7	Negative	Invalid OTP Validation	Invalid OTP validation handled successfully	Passed

END-TO-END FLOW VALIDATION

The End-To-End automation validates the complete business workflow including:

- User Login Validation
- Pharmacy Page Navigation
- Brand Filter Application
- Price Filter Application
- Product Selection Validation
- Add Product To Basket Validation
- Basket Navigation Validation
- Checkout Navigation Validation
- Payment Page Verification

POSITIVE TEST SCENARIOS

Positive Test Scenario 1 – Brand Filter Validation

- Validates successful navigation to the BigBasket Pharmacy & Wellness page
- Verifies pharmacy module loading successfully

- Selects “Dettol” brand filter dynamically
 - Verifies successful brand filter application
-

Positive Test Scenario 2 – Price Filter Validation

- Validates price filter functionality
 - Selects product price range dynamically
 - Verifies successful price filter application
 - Ensures filtered pharmacy products are displayed correctly
-

Positive Test Scenario 3 – Add Product To Basket Validation

- Validates add-to-basket functionality
 - Selects pharmacy product dynamically
 - Adds selected product to basket successfully
 - Verifies successful basket update validation
-

Positive Test Scenario 4 – Basket Navigation Validation

- Validates basket opening functionality
 - Opens basket page successfully
 - Verifies successful basket page navigation
 - Ensures added products are displayed correctly inside basket
-

NEGATIVE TEST SCENARIOS

Negative Test Scenario 1 – Invalid Mobile Number Validation

- Enters invalid mobile number value
 - Validates application error handling
 - Verifies invalid mobile number validation successfully
 - Ensures unauthorized login prevention
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Negative Test Scenario 2 – Invalid OTP Validation

- Enters valid mobile number
 - Navigates to OTP verification page
 - Verifies invalid OTP validation handling
 - Ensures user login restricted for invalid OTP values
-

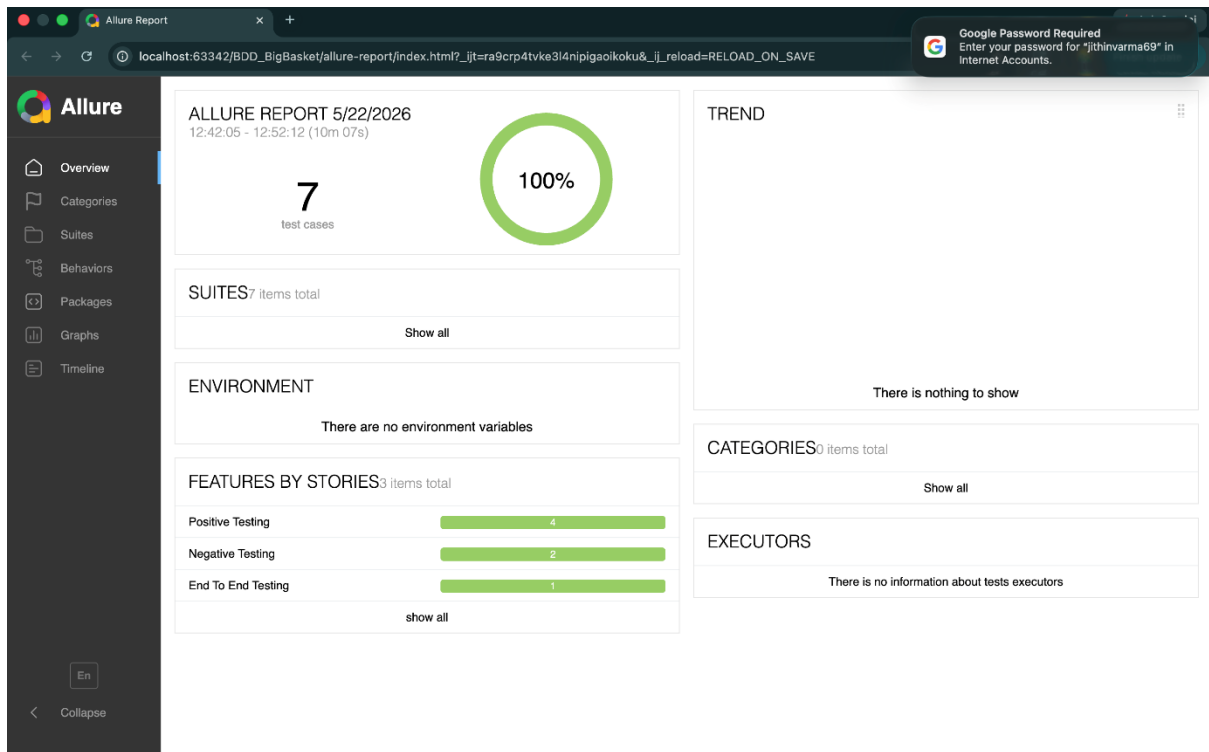
BDD IMPLEMENTATION

The framework follows Behavior Driven Development using:

- Feature Files
 - Step Definition Files
 - Gherkin Syntax
 - Given-When-Then Structure
 - Reusable Step Implementations
 - Scenario-Based Automation Execution
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ALLURE REPORTS

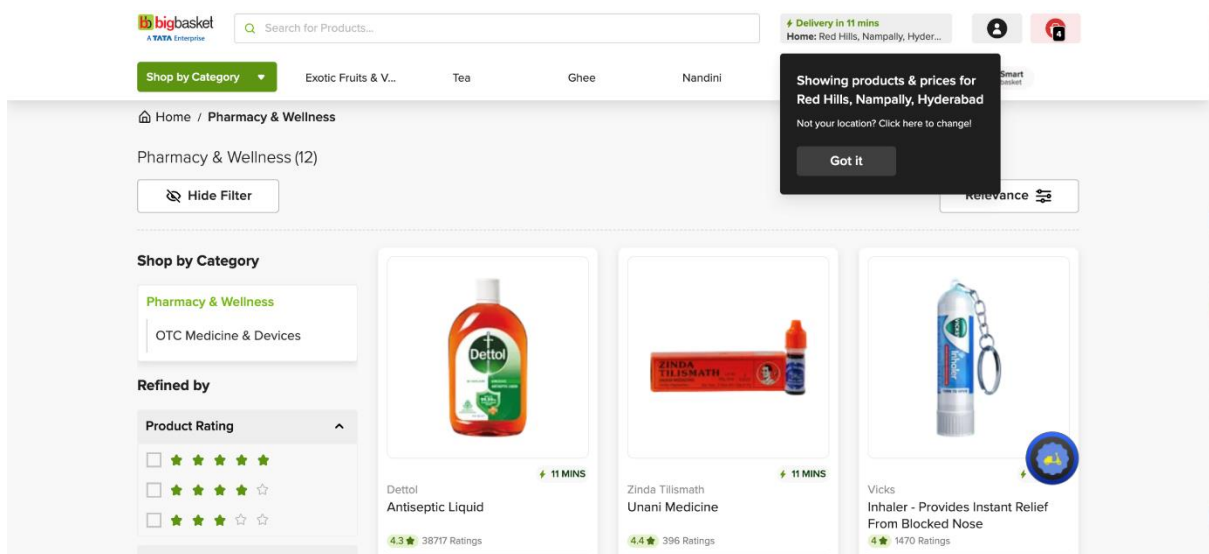
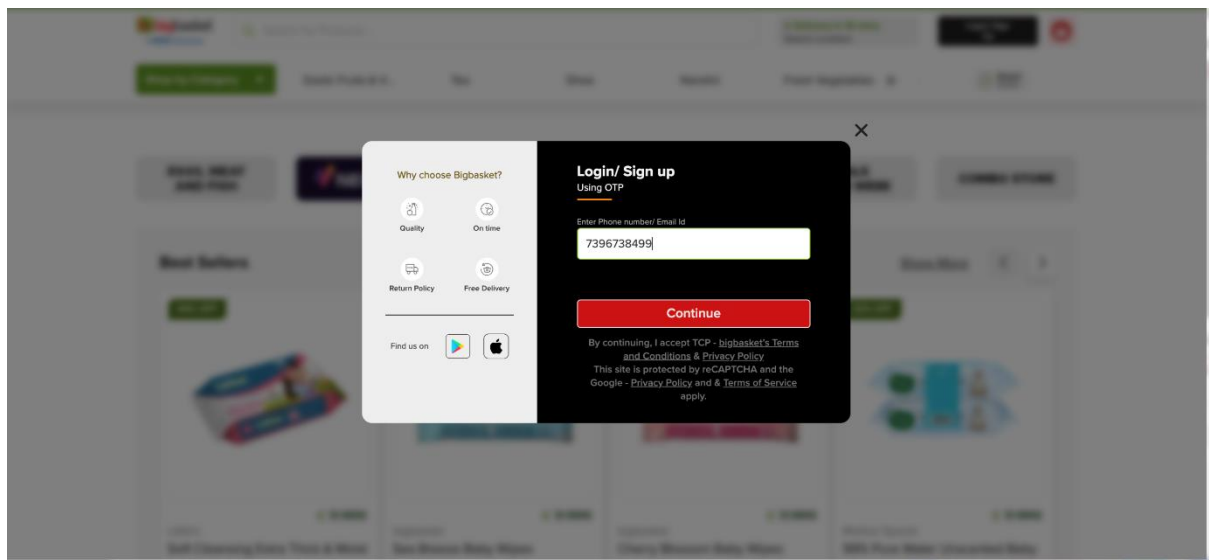
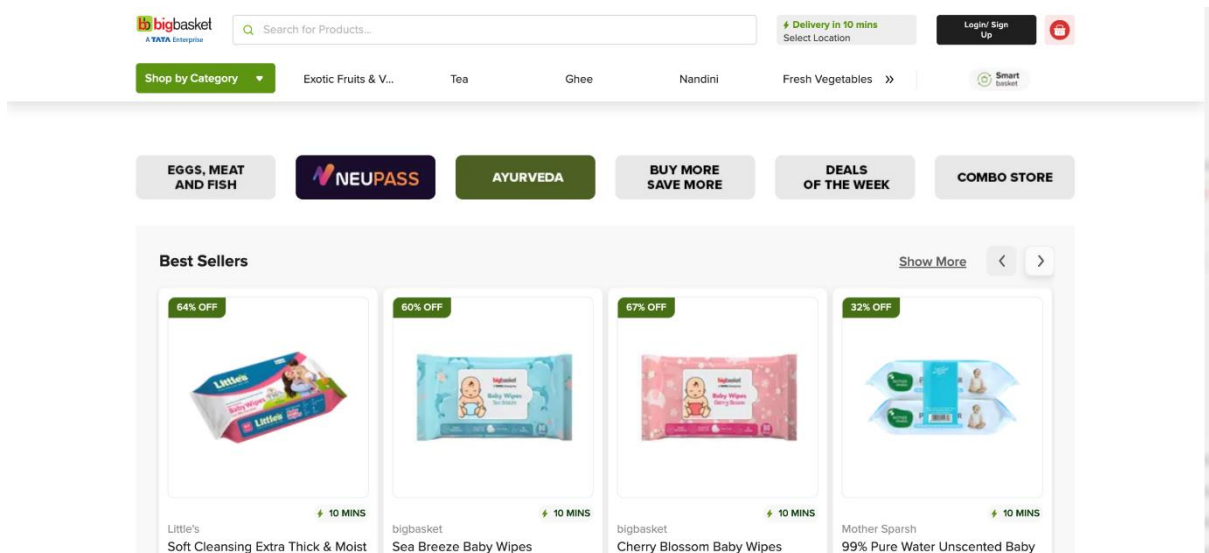
- Integrated Allure Reports with Behave BDD
- Generates professional execution reports
- Displays passed and failed scenario status
- Captures screenshots during execution
- Helps debugging and tracking failures
- Provides detailed execution summary
- Displays assertion validation results
- Generates real-time automation execution analytics
- Improves reporting quality and execution transparency



TEST OUTPUTS

The framework automatically captures screenshots during important execution stages including user login validation, pharmacy page navigation, brand filter application, price filter validation, product selection, add-to-basket functionality, basket page validation, checkout workflow navigation, and payment page verification.

Detailed execution logs are generated using the Python logging framework for debugging, validation tracking, execution monitoring, assertion verification, and automation flow analysis. The logging framework helps in identifying failures, tracking execution flow, monitoring scenario validations, and improving automation maintenance and debugging efficiency.



Delivery in 11 mins
Home: Red Hills, Nampally, Hyder...
4

Shop by Category
Exotic Fruits & V...
Tea
Ghee
Nandini
Fresh Vegetables
Smart basket

Home / Pharmacy & Wellness

Pharmacy & Wellness (12)

Filters
Clear
Dettol X 4 ★ ★ ★ ★ X

Refined by
Product Rating
Clear
4.3 ★ ★ ★ ★ ★
Brands
Clear
Dettol

Dettol Antiseptic Liquid
11 MINS
4.3 ★ 38717 Ratings

Dettol Antiseptic Liquid
11 MINS
4.3 ★ 38717 Ratings

Dettol Antiseptic Liquid
11 MINS
4.3 ★ 38717 Ratings

Delivery in 11 mins
Home: Red Hills, Nampally, Hyder...
4

Shop by Category
Exotic Fruits & V...
Tea
Ghee
Nandini

Your Basket

Delivery in 11 mins

Showing products & prices for Red Hills, Nampally, Hyderabad
Not your location? Click here to change!
Got it

4 Products

Items	Quantity	Sub-total
Antiseptic Liquid ₹364.19	- 5 +	₹1820.95
Delete Save for later		

Subtotal: ₹2486.95

Choose delivery type

☒ Now 11 mins
☐ Later 3 hrs

☒ Delivery Address
Home - 5-64 kr Red Hills, Nampally
☒ Payment Options
Pay Order amount by selecting any payment mode

Credit / Debit Card

Netbanking
UPI
Wallets
Pluixee card
Non-Food item(s) worth 2486.95rs in your cart.Pay for non-food items by adding this amount to your bbWallet(My Account -> My Wallet -> Fund Wallet)
Pay on delivery

Enter Credit / Debit card details

Card number
Expiry
CVV
Pay ₹2,492.95

☒ Apply Voucher
0 vouchers are available
View all

Order Summary

Basket Value ₹2486.95
Delivery & Handling Charges ₹6
Show Details
Get your bbNow order in a bag Remove FREE
Use bbWallet ₹0
Balance: ₹0
Total Amount Payable ₹2492.95

TEST REPORT

- ❑ Total Scenarios Executed: 7
- ❑ Scenarios Passed: 7
- ❑ Scenarios Failed: 0
- ❑ Automation Accuracy: 100%

All BDD automation scenarios executed successfully without any execution failures. The framework successfully validated the complete BigBasket Pharmacy & Wellness workflow including positive testing, negative testing, end-to-end testing, assertion validation, screenshot capture, professional reporting, and detailed execution tracking using Behave BDD framework.

CONCLUSION

The Selenium Behave BDD automation framework successfully validates the BigBasket Pharmacy & Wellness application using Behavior Driven Development methodology.

The framework is implemented using:

- Selenium WebDriver
- Behave BDD Framework
- Gherkin Feature Files
- Step Definition Files
- Page Object Model (POM)
- Allure Reporting
- Data-Driven Testing
- Screenshot Capture
- Assertion Validations
- Logging Framework

The framework validates complete business workflows including:

- User Login Validation
- Pharmacy Page Navigation
- Brand Filter Application
- Price Filter Validation

- Product Selection Validation
- Add Product To Basket Validation
- Basket Navigation Validation
- Checkout Navigation Validation
- Payment Page Verification
- Invalid Mobile Number Validation
- Invalid OTP Validation

The implementation of Feature Files and Given-When-Then structure improves readability, maintainability, reusability, scalability, and collaboration between technical and non-technical users.

The framework also includes:

- Professional Allure Reports
- Screenshot Evidence
- Detailed Execution Logs
- Positive and Negative Testing
- Reusable Automation Components
- Assertion Verification
- Step-By-Step Scenario Execution Flow
- Real-Time Execution Tracking

All scenarios executed successfully with accurate validations, detailed reporting, reusable automation components, screenshot evidence, assertion tracking, and professional execution logs.

Overall, the project demonstrates a professional real-world BDD automation framework with scalable architecture, reusable components, maintainable automation design, and reliable automated testing practices for real-world e-commerce web application testing.